$\mathcal{K}_{0^+}^{\#1}$ $\mathcal{K}_{0^-}^{\#1\,\alpha\beta\chi}$

$\mathcal{K}_{0^+}^{\#1}\dagger$	$\frac{1}{8}(-3\,\Upsilon_1\,k^2-12\,\overset{(2)}{\kappa}_3)$ | 0
$\mathcal{K}_{0^-}^{\#1}\dagger^{\alpha\beta\chi}$	0 | $3\,k^2\,\overset{(4)}{\kappa}_{15}$

$\mathcal{K}_{1^+}^{\#1\,\alpha\beta}$ $\mathcal{K}_{1^+}^{\#2\,\alpha\beta}$ $\mathcal{K}_{1^-}^{\#1\,\alpha}$ $\mathcal{K}_{1^-}^{\#2\,\alpha}$

$\mathcal{K}_{1^+}^{\#1}\dagger^{\alpha\beta}$	$-\frac{\Upsilon_2\,k^2}{8}$	$\frac{\Upsilon_2\,k^2}{4\sqrt{2}}$	0	0
$\mathcal{K}_{1^+}^{\#2}\dagger^{\alpha\beta}$	$\frac{\Upsilon_2\,k^2}{4\sqrt{2}}$	$-\frac{\Upsilon_2\,k^2}{4}$	0	0
$\mathcal{K}_{1^-}^{\#1}\dagger^{\alpha}$	0	0	$-k^2\,\overset{(4)}{\kappa}_2-\overset{(2)}{\kappa}_3$	$\frac{k^2\,\overset{(4)}{\kappa}_2+\overset{(2)}{\kappa}_3}{\sqrt{2}}$
$\mathcal{K}_{1^-}^{\#2}\dagger^{\alpha}$	0	0	$\frac{k^2\,\overset{(4)}{\kappa}_2+\overset{(2)}{\kappa}_3}{\sqrt{2}}$	$\frac{1}{2}(-k^2\,\overset{(4)}{\kappa}_2-\overset{(2)}{\kappa}_3)$

$\mathcal{K}_{2^+}^{\#1\,\alpha\beta}$ $\mathcal{K}_{2^-}^{\#1\,\alpha\beta\chi}$

$\mathcal{K}_{2^+}^{\#1}\dagger^{\alpha\beta}$	$\frac{9\,\Upsilon_2\,k^2}{8}$	0
$\mathcal{K}_{2^-}^{\#1}\dagger^{\alpha\beta\chi}$	0	0

$\mathcal{J}_{0^+}^{\#1}$ $\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi}$

$\mathcal{J}_{0^+}^{\#1}\dagger$	$\frac{1}{\text{Det}(0^+)}$	0
$\mathcal{J}_{0^-}^{\#1}\dagger^{\alpha\beta\chi}$	0	$\frac{1}{3\,k^2\,\overset{(4)}{\kappa}_{15}}$

$\mathcal{J}_{1^+}^{\#1\,\alpha\beta}$ $\mathcal{J}_{1^+}^{\#2\,\alpha\beta}$ $\mathcal{J}_{1^-}^{\#1\,\alpha}$ $\mathcal{J}_{1^-}^{\#2\,\alpha}$

$\mathcal{J}_{1^+}^{\#1}\dagger^{\alpha\beta}$	$\frac{1}{3\,\text{Det}(1^+)}$	$-\frac{\sqrt{2}}{3\,\text{Det}(1^+)}$	0	0
$\mathcal{J}_{1^+}^{\#2}\dagger^{\alpha\beta}$	$-\frac{\sqrt{2}}{3\,\text{Det}(1^+)}$	$\frac{2}{3\,\text{Det}(1^+)}$	0	0
$\mathcal{J}_{1^-}^{\#1}\dagger^{\alpha}$	0	0	$\frac{2}{3\,\text{Det}(1^-)}$	$-\frac{\sqrt{2}}{3\,\text{Det}(1^-)}$
$\mathcal{J}_{1^-}^{\#2}\dagger^{\alpha}$	0	0	$-\frac{\sqrt{2}}{3\,\text{Det}(1^-)}$	$\frac{1}{3\,\text{Det}(1^-)}$

$\mathcal{J}_{2^+}^{\#1\,\alpha\beta}$ $\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi}$

$\mathcal{J}_{2^+}^{\#1}\dagger^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0
$\mathcal{J}_{2^-}^{\#1}\dagger^{\alpha\beta\chi}$	0	0

Abbreviations used in matrices

$$\Upsilon_1 == 4\,\overset{(4)}{\kappa}_1-6\,\overset{(4)}{\kappa}_{15}+4\,\overset{(4)}{\kappa}_2+3\,\overset{(4)}{\kappa}_3\;\&\&\;\Upsilon_2 == 2\,\overset{(4)}{\kappa}_{15}-\overset{(4)}{\kappa}_3\;\&\&$$

$$\text{Det}(0^+) == -\frac{3}{8}(k^2(4\,\overset{(4)}{\kappa}_1-6\,\overset{(4)}{\kappa}_{15}+4\,\overset{(4)}{\kappa}_2+3\,\overset{(4)}{\kappa}_3)+4\,\overset{(2)}{\kappa}_3)\;\&\&$$

$$\text{Det}(1^+) == \frac{3}{8}k^2(-2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_3)\;\&\&\;\text{Det}(2^+) == \frac{9}{8}k^2(2\,\overset{(4)}{\kappa}_{15}-\overset{(4)}{\kappa}_3)$$

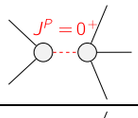
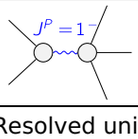
Lagrangian

$$\overset{(2)}{\kappa}_3\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}^{\chi}{}_{\beta\chi}+\overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+\overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+\overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta}-$$

$$3\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta}-\frac{1}{2}\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta}-\frac{9}{2}\,\overset{(4)}{\kappa}_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta}+\frac{5}{4}\,\overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta}-$$

$$\frac{3}{2}\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}+\frac{1}{4}\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}+\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}-2\,\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{1^-}^{\#1\,\alpha}+2\;\mathcal{J}_{1^-}^{\#2\,\alpha}==0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi}+\partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta}==3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
$2\;\mathcal{J}_{1^+}^{\#1\,\alpha\beta}+\mathcal{J}_{1^+}^{\#2\,\alpha\beta}==0$	3	$\partial_\chi\mathcal{J}^{\alpha\beta\chi}+\partial_\chi\mathcal{J}^{\chi\alpha\beta}==\partial_\chi\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi}==0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta}+$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi}+3\;\eta^{\beta\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\;\eta^{\alpha\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon}+3\;\eta^{\beta\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta}==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta}+$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}+3\;\eta^{\alpha\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\;\eta^{\beta\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon}+3\;\eta^{\alpha\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	11	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$-\frac{4\,\overset{(2)}{\kappa}_3}{4\,\overset{(4)}{\kappa}_1-6\,\overset{(4)}{\kappa}_{15}+4\,\overset{(4)}{\kappa}_2+3\,\overset{(4)}{\kappa}_3}$	$-\frac{8}{3(4\,\overset{(4)}{\kappa}_1-6\,\overset{(4)}{\kappa}_{15}+4\,\overset{(4)}{\kappa}_2+3\,\overset{(4)}{\kappa}_3)}$
	3	$-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_2}$	$\frac{2}{3\,\overset{(4)}{\kappa}_2}$
Resolved unitarity condition(s):		(Demonstrably impossible)	